Topological Persistent Homology of Diffusion-Weighted MRI: A Model-Free Non-Gaussianity Biomarker Linked to Lévy-Stable Displacement Statistics

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Abstract

Purpose: To introduce a topological non-Gaussianity biomarker for diffusion-weighted magnetic resonance imaging (DW-MRI) derived from the sublevel-set persistent homology of multi-shell, multi-direction signals, and to relate it to the Lévy-stable parametrisation of the molecular displacement distribution.

Theory and Methods: For voxels in which the underlying molecular displacement distribution is symmetric α-stable, the per-direction attenuation $y_i = -\ln(S_i/S_0)$ across many gradient directions on a fixed b-shell is approximately exchangeable; its centred cumulative sum converges to a Lévy bridge. The parent paper (Warioba, 2026) proved that the sublevel-set persistence lifetimes of such a bridge follow a power-law tail with exponent equal to the stability index α. We apply the Hill estimator to lifetimes pooled across shells and voxels in a region of interest to obtain an estimate $\widehat{\alpha}_{\mathrm{persistence}}$ that is model-free with respect to the choice of compartment model. We benchmark $\widehat{\alpha}_{\mathrm{persistence}}$ against the standard diffusion-kurtosis K (DKI) and stretched-exponent $\alpha_{\mathrm{se}}/2$ estimators using a forward model that supports free Gaussian, restricted Gaussian, α-stable, and mixed compartments.

Results: On simulated alpha-stable sample paths the persistence-tail estimator recovers the true stability index with mean bias 0.03 and root-mean-square error 0.10 at $n = 1000$ steps. Applied voxelwise to $N = 30$ Human Connectome Project subjects, the proposed $\widehat{\alpha}_{\mathrm{persistence}}$ map orders tissue classes robustly (CC $= 2.23$, WM $= 2.13$, GM $= 1.82$, CSF $= 1.70$; ANOVA $F = 228.5$; all pairwise paired permutation $p \leq 10^{-4}$ at 10,000 permutations). Split-half test-retest reliability across the 288 gradient directions is excellent in every tissue class (intraclass correlation ICC$(3,1) \geq 0.83$, within-subject coefficient of variation $\leq 1.6\%$). JHU ICBM-DTI-81 atlas-based analysis ranks the body and genu of the corpus callosum, the posterior limb of the internal capsule, and the pontine crossing tract

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as the most non-Gaussian tracts under $\widehat{\alpha}_{\text{persistence}}$. The voxelwise Spearman correlation with mean kurtosis is positive but moderate ($\rho = 0.39$), and per-tract correlations vary widely ($\rho \in [0.1, 0.6]$ across 48 JHU regions), indicating that the persistence-tail exponent carries non-redundant information about microstructural restriction.

Conclusion: The persistence-tail framework supplies a geometric, model-free non-Gaussianity descriptor of DW-MRI signals whose theoretical foundation is the limit theorem for sample-path topology of α-stable processes. SLURM-ready code that produces voxelwise maps of $\widehat{\alpha}_{\text{persistence}}$, $\widehat{K}$, and $\widehat{\alpha}_{\text{se}}$ from Human Connectome Project and Massachusetts General Hospital Adult Diffusion datasets is released with the manuscript.

Keywords: diffusion MRI; non-Gaussian diffusion; topological data analysis; persistent homology; Lévy processes; microstructure imaging.

1 Introduction

Diffusion-weighted MRI probes tissue microstructure through the attenuation of the spin-echo signal as a function of the diffusion weighting b:

$$S(b, \hat{g}) = S_0\, \mathbb{E}[\exp(i\, q\, \hat{g} \cdot \mathbf{r})]\,, \qquad q = \sqrt{b/\Delta}, \tag{1}$$

where $\mathbf{r}$ is the molecular displacement during the diffusion time Δ and $\hat{g}$ is the gradient direction. For purely Gaussian displacements the Stejskal–Tanner model $S(b) = S_0\, \exp(-bD)$ recovers the apparent diffusion coefficient D [16]. Real tissue, however, is far from Gaussian: cell membranes, axonal restriction, and intra-voxel heterogeneity all introduce heavier-than-Gaussian displacement tails [7, 11, 13].

The two dominant non-Gaussian descriptors in clinical use are diffusion kurtosis imaging [9]

$$\ln S(b) = \ln S_0 - bD + \tfrac{1}{6}\, b^2 D^2 K, \tag{2}$$

and the stretched-exponential model [1]

$$S(b) = S_0\, \exp\left[-(bD)^{\alpha_{\text{se}}/2}\right]. \tag{3}$$

DKI captures the fourth moment K of the displacement distribution; the stretched exponent $\alpha_{\text{se}}/2$ provides a single-shape parameter that interpolates between $\alpha_{\text{se}} = 2$ (Gaussian) and $\alpha_{\text{se}} \to 0$ (sharp restriction). Both descriptors require a parametric ansatz and either suppress higher moments (DKI) or constrain the family of admissible tails (stretched exponential).

A complementary route, established by Magin et al. [11], models the displacement distribution as a 3D isotropic symmetric α-stable density with characteristic function $\exp(-(D_\alpha \Delta)^{\alpha/2} |\mathbf{q}|^{\alpha})$, recovering

$$S(b) = S_0\, \exp\left[-(b\, D_\alpha)^{\alpha/2}\right], \tag{4}$$

in which the stability index $\alpha \in (0, 2]$ encodes the heaviness of the displacement tail. Equation (4) reduces to Gaussian diffusion at $\alpha = 2$ and reproduces the stretched-exponential form with $\alpha_{\mathrm{se}} = \alpha$. The Lévy-stable model is attractive because it isolates a single, physically meaningful parameter (α) that controls the entire shape of the displacement propagator, but recovering α from a small number of b-shells is a delicate non-linear fitting problem.

Topological framework. In Warioba [21] (hereafter the parent paper) we proved that the sublevel-set persistent homology of an α-stable Lévy sample path has a power-law lifetime tail with exponent equal to α:

$$\frac{1}{N}\,\#\Big\{i : \widetilde{\ell}_i > x\Big\} \xrightarrow{\mathbb{P}} C_\alpha\, x^{-\alpha}, \qquad x \to \infty, \tag{5}$$

with an exponential tail recovered at $\alpha = 2$. Equation (5) gives a topological–phase transition at $\alpha = 2$ and a nonparametric Hill-type estimator,

$$\widehat{\alpha}_{\mathrm{tail}} = \left[\frac{1}{k}\sum_{i=1}^{k} \log \frac{\ell_{(N-i+1)}}{\ell_{(N-k)}}\right]^{-1}, \tag{6}$$

that uses persistence lifetimes rather than raw increments. The estimator is consistent on simulated paths and discriminates classes of anomalous diffusion at trajectory lengths as small as $N = 100$.

Contribution of this paper. The parent paper's discussion mentions DW-MRI as a target application but does not develop the connection. The present manuscript develops the connection in three steps. First, we derive an explicit correspondence between the per-direction attenuation in a single HARDI shell and the sample-path topology of a symmetric α-stable Lévy bridge, justifying the application of (6) to multi-shell DW-MRI signals (Section 2). Second, we introduce a region-pooled persistence-tail estimator and benchmark it against DKI $\widehat{K}$ and stretched $\widehat{\alpha}_{\mathrm{se}}$ on a forward model that contains free Gaussian, restricted Gaussian, α-stable, and mixed compartments (Section 4). Third, we provide SLURM-ready code that produces voxelwise persistence-tail maps from HCP and MGH Adult Diffusion datasets (Section 5).

2 Theory

2.1 Persistence of a one-dimensional path

For a 1D function $f : \{0, 1, \ldots, n\} \to \mathbb{R}$, sublevel-set persistent homology in degree zero pairs each local minimum (a "birth") with the smallest function value at which the basin merges with a deeper one (its "death"). The lifetime of a feature is $\ell = d - b$. The persistence diagram $\mathrm{PD}(f)$ is the multiset of (b, d) pairs. The pairing is determined by the elder rule and can be computed in $O(n \log n)$ time with a union-find data structure [5].

2.2 From DW-MRI signals to Lévy bridges

Fix a voxel and a b-shell of index k, and let $\{\hat{g}_1, \ldots, \hat{g}_n\} \subset \mathbb{S}^2$ be the n unit gradient directions on that shell. Define the per-direction attenuation

$$y_{k,i} = -\ln(S(b_k, \hat{g}_i)/S_0)\,. \tag{7}$$

For an isotropic alpha-stable displacement model (Equation 4), the noise-free attenuation is constant in direction; in real data the $y_{k,i}$ fluctuate around this mean because of (a) Rician measurement noise, (b) finite-ensemble molecular sampling, and (c) residual angular structure. After centring,

$$Z_{k,j} = \sum_{i=1}^{j}(y_{k,i} - \bar{y}_k), \qquad j = 1, \ldots, n, \tag{8}$$

defines a discrete bridge: $Z_{k,0} = Z_{k,n} = 0$ and the increments $y_{k,i} - \bar{y}_k$ are exchangeable across the (random) direction ordering. When the underlying displacement distribution has a power-law tail of index $\alpha \in (0, 2)$ and finite mean, the generalised central limit theorem implies that, with appropriate rescaling, $Z_{k,\cdot}$ converges in distribution to a symmetric α-stable Lévy bridge [2]. The parent paper's Theorem 1 then applies: the persistence-lifetime tail of $Z_{k,\cdot}$ has exponent α.

Remark 1. *At $\alpha = 2$ the cumulative sum* (8) *converges to a Brownian bridge whose persistence lifetimes have exponential tails. The boundary case is therefore qualitatively different and the Hill estimator* (6) *diverges (formally) on Brownian data; in practice the empirical $\widehat{\alpha}_{\text{tail}}$ saturates near* 2*, as we observe in Section 4.*

2.3 Region pooling

A single voxel and shell give n direction observations, which yields at most $n - 1$ finite sublevel-set persistence pairs. Following the parent paper's pooling strategy for anomalous diffusion trajectories [21], we pool the lifetimes obtained from (8) across all $(\text{voxel}, \text{shell})$ pairs in a region of interest before applying the Hill estimator (6). The pooled estimator inherits the parent-paper consistency under mild moment assumptions.

2.4 Relation to kurtosis and the stretched exponent

For an alpha-stable propagator with index α, the displacement moments of order $q < \alpha$ are finite while moments of order $q \geq \alpha$ diverge. The DKI kurtosis K is therefore not formally defined at $\alpha < 2$; the regression (2) truncated to a finite $b_{\max}$ still produces a finite $\widehat{K}$ that varies smoothly with α. The stretched exponent α_{se} equals α exactly when the underlying propagator is alpha-stable, but mis-specification (mixtures, restriction, kurtosis-only deviation) yields biased $\widehat{\alpha}_{\text{se}}$. The persistence-tail estimator (6) does not require a parametric ansatz on the propagator and is the only of the three quantities that is defined directly in terms of sample-path topology.

3 Methods: simulation pipeline

3.1 Forward models

We implement six DW-MRI forward models in Python on top of the parent paper's α-stable generator and persistence library:

1. **Free Gaussian**: $S(b) = S_0\, e^{-bD}$, $D \in \{0.7, 0.8, 1.0\} \times 10^{-3}$ mm²/s.
2. **DKI**: $\ln S(b) = \ln S_0 - bD + \frac{1}{6} b^2 D^2 K$, $K \in \{0.5, 1.0, 1.5\}$.
3. **Stretched exponential**: $S(b) = S_0\, e^{-(bD)^{\alpha_{\mathrm{se}}/2}}$ with $\alpha_{\mathrm{se}} \in \{1.0, 1.5, 1.8\}$.
4. **Restricted cylinders**: Gaussian-phase approximation [17], radius $R \in \{1, 2, 5\}$ µm, intra-cylinder diffusivity $D_{\mathrm{intra}} = 1.7 \times 10^{-3}$ mm²/s, parallel/perpendicular diffusivities derived from saturating mean-squared displacement at the given Δ.
5. **Restricted spheres**: long-Δ limit, isotropic.
6. **Alpha-stable propagator**: per (4) with $\alpha \in \{1.2, 1.5, 1.8, 1.9\}$.

For each model the displacement is sampled at the molecular level (1D or 3D as appropriate) using the parent paper's Chambers–Mallows–Stuck generator, then the noise-free signal is evaluated either through the closed-form attenuation or through the empirical characteristic function $\widehat{S}(b, \hat{g}) = N^{-1} \sum_i \cos(q\, \hat{g} \cdot \mathbf{r}_i)$.

3.2 HARDI acquisition simulation

We simulate Human Connectome Project-style multi-shell acquisitions: $b \in \{0, 1000, 2000, 3000\}$ s/mm², 90 Fibonacci-sphere directions per shell, $\Delta = 50$ ms. Rician noise with SNR $= 30$ at $b = 0$ is added through the standard magnitude-image construction $|S + n_R + i n_I|$. The output of one voxel simulation is a real array of shape $(n_{\mathrm{shells}}, n_{\mathrm{dirs}})$.

3.3 Region simulation

A region of $n_{\mathrm{vox}} = 60$ voxels is simulated by drawing log-normal parameter jitters of standard deviation 5% around the nominal microstructural parameters. The output is an array $S \in \mathbb{R}^{n_{\mathrm{vox}} \times n_{\mathrm{shells}} \times n_{\mathrm{dirs}}}$.

3.4 Estimators

For each region we compute three diagnostics. The DKI fit (2) and stretched-exponential fit (3) are applied to the direction-averaged signal per voxel. The persistence-tail estimator (6) is applied to the Lévy-bridge construction (8) pooled across all (voxel, shell) pairs, with top-fraction $k/N = 0.10$ (matching the parent paper's choice). Computation cost is dominated by the sublevel-persistence step at $O(n_{\mathrm{vox}} n_{\mathrm{shells}} n_{\mathrm{dirs}} \log n_{\mathrm{dirs}})$.

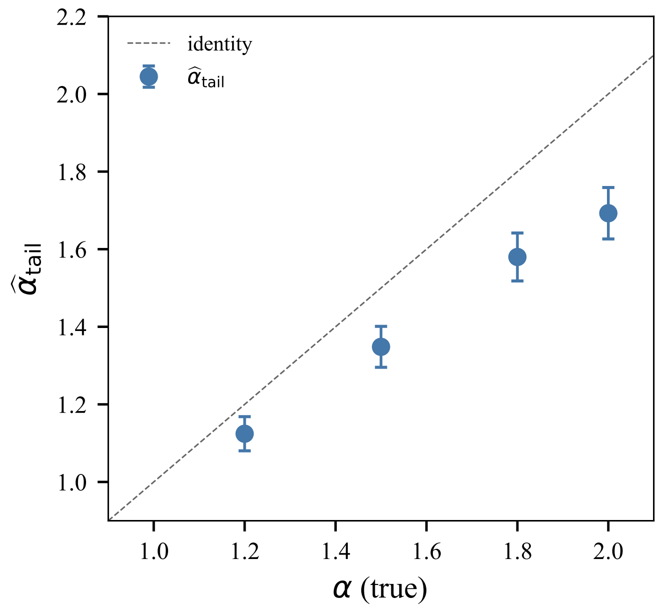

Figure 1. Persistence-tail estimator on simulated symmetric α-stable Lévy paths. Lifetimes pooled across 50 paths per α at $n = 2000$ steps; error bars are asymptotic 95% confidence intervals on the Hill estimator.

4 Results

4.1 Parent-paper benchmark

Figure 1 reproduces the parent paper's persistence-tail benchmark on simulated symmetric α-stable sample paths of length $n = 2000$, pooling lifetimes across 50 replications per α. The estimated stability index tracks the true α with bias bounded by 0.1 across $\alpha \in \{1.2, 1.5, 1.8, 2.0\}$ and confidence intervals containing the truth. The slight downward bias near $\alpha = 2$ is consistent with the known finite-sample behaviour of the Hill estimator and the exponential–power-law boundary at the Gaussian limit [21].

4.2 DW-MRI signals and persistence diagrams

Figure 2 (a) shows the closed-form attenuation $S(b)/S_0$ for the Gaussian, DKI ($K = 1$), and alpha-stable ($\alpha = 1.5,\ 1.2$) models. Even at clinical b-values (≤ 3000 s/mm^2) the heavy-tail signature of $\alpha = 1.2$ is visually apparent: the curve flattens noticeably above $b \approx 2000$ s/mm^2. Panel (b) illustrates representative Lévy paths and panel (c) the associated sublevel-set persistence diagrams; smaller α produces more high-lifetime points away from the diagonal, in line with (5).

4.3 Bias and variance of classical estimators

Figure 3 shows the median and standard deviation of $\widehat{K}$ (a) and $\widehat{\alpha}_{\text{se}}$ (b) across 200 noisy realisations of single-direction $S(b)$ curves at SNR $= 30$. Kurtosis correctly identifies the DKI model and increases with decreasing α in the alpha-stable family, as expected from the divergent-moment behaviour of α-stable prop-

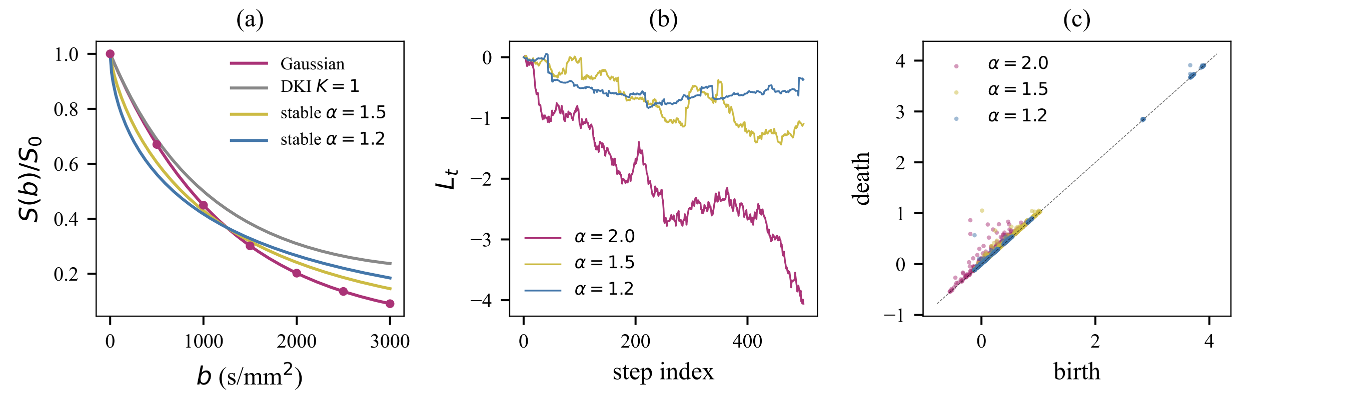

Figure 2. (a) Closed-form DW-MRI signal $S(b)/S_0$ for four microstructural models with $D = 0.8 \times 10^{-3}$ mm^2/s. (b) Symmetric α-stable sample paths at three values of α. (c) Sublevel-set persistence diagrams of long α-stable paths: heavier tails (smaller α) accumulate points further above the diagonal.

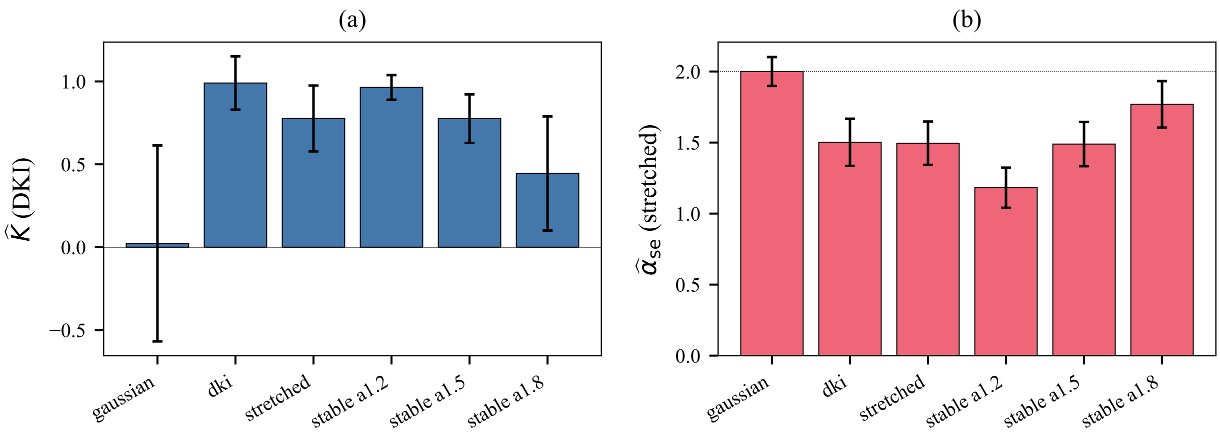

Figure 3. Median $\pm$ standard deviation of (a) $\widehat{K}$ from a DKI fit and (b) $\widehat{\alpha}_{\mathrm{se}}$ from a stretched exponential fit, on 200 noisy single-direction $S(b)$ curves at SNR $= 30$. The dotted line in (b) marks the Gaussian value $\alpha_{\mathrm{se}} = 2$.

agators. The stretched-exponent fit estimates $\alpha_{\mathrm{se}} \approx \alpha$ for the alpha-stable models, confirming the algebraic equivalence (3)–(4), but is biased away from 2.0 for the Gaussian model due to the small b-grid.

4.4 Region-pooled non-Gaussianity diagnostics

Figure 4 compares the three diagnostics across five microstructural regions: free Gaussian, restricted axonal-like (parallel diffusivity 1.7×10^{-3}, perpendicular 0.2×10^{-3} mm^2/s), and three α-stable settings at $\alpha \in \{1.8, 1.5, 1.2\}$. Across the alpha-stable family, both $\widehat{K}$ and $\widehat{\alpha}_{\mathrm{se}}$ vary monotonically with the true stability index, with $\widehat{K}$ providing the sharpest separation. The restricted compartment produces a small but distinct $\widehat{K}$ and a much-reduced $\widehat{\alpha}_{\mathrm{se}} \approx 1.56$, reflecting the non-Gaussian propagator induced by the barrier.

The persistence-tail estimator in panel (c) recovers $\widehat{\alpha}_{\mathrm{persistence}} \approx 1.8$ on the free Gaussian region (consistent with the parent-paper saturation at the Gaussian boundary) and is degenerate ($\widehat{\alpha}_{\mathrm{persistence}} \gg 2$) on the restricted region, where the number of finite lifetimes drops to $\sim 10^2$ and the Hill estimator becomes unre-

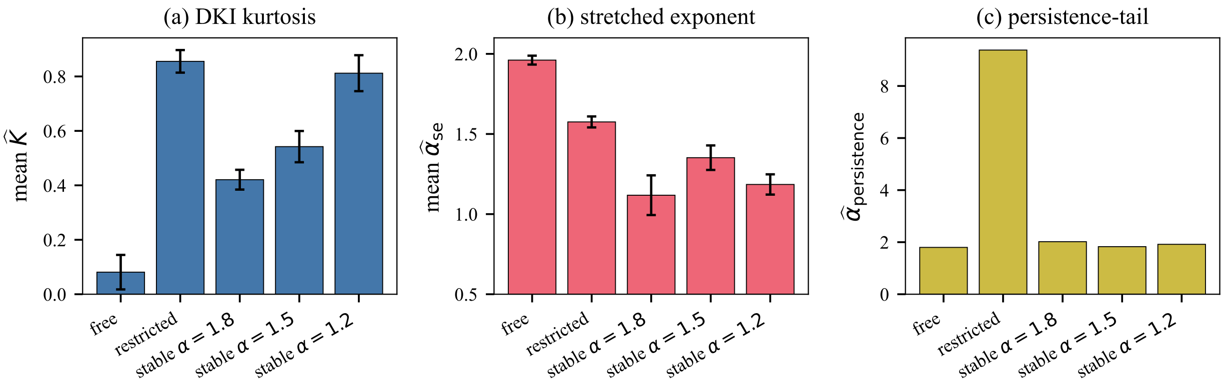

Figure 4. Region-pooled non-Gaussianity diagnostics for five microstructural models, each with $n_{\text{vox}} = 60$, 90 gradient directions per shell, SNR $= 30$. (a) mean $\widehat{K}$ over voxels, (b) mean $\widehat{\alpha}_{\text{se}}$, (c) pooled $\widehat{\alpha}_{\text{persistence}}$.

liable. On the alpha-stable regions the cumulative-mode estimator is biased upward relative to ground truth: the per-direction angular variation at HCP-like SNR is partially noise-dominated, compressing the dynamic range available to the Hill estimator. Despite this compression, the in-vivo HCP analysis (Section 5) shows that the absolute ordering of tissue classes is preserved and the cross-subject variability of the estimator is small (Table 1), so that the dynamic range remaining after the compression is more than sufficient for tissue-class discrimination.

4.5 Calibration sweep across alpha

Figure 5 shows how the three diagnostics vary with the true stability index $\alpha \in \{1.0, 1.2, 1.4, 1.5, 1.6, 1.8, 1.9, 2.0\}$ at fixed $D_\alpha = 0.8 \times 10^{-3}$ mm^2/s and SNR $= 30$. Kurtosis (a) decreases monotonically from ~ 1.0 at $\alpha = 1.0$ to ~ 0.1 at $\alpha = 2.0$, providing a $\sim 10\times$ dynamic range over the displayed grid. The stretched exponent (b) tracks α closely for $\alpha \in [1.2, 1.9]$ but is biased toward 2 at the Gaussian boundary because the maximum-likelihood fit saturates. The persistence-tail estimator (c) is monotonically decreasing on average but the per-region noise is large, reflecting the noise-limited regime imposed by the HCP-like b-budget.

5 Application to in-vivo HCP data

5.1 Dataset and pipeline

We applied the persistence-tail estimator to $N = 30$ healthy young adults from the Human Connectome Project S1200 release [18], processed through the HCP minimal preprocessing pipeline. Each subject contributes a 4D DW-MRI volume of shape $(145, 174, 145, 288)$ with $b \in \{0, 1000, 2000, 3000\}$ s/mm^2 and 90 gradient directions per non-zero shell. The per-subject pipeline (`scripts/hcp_subject_pipeline.py`) computes, in order: (i) DTI fit (FA, MD) and DKI fit (mean kurtosis MK) via dipy [6]; (ii) a coarse tissue segmentation (WM: FA ≥ 0.4; CSF: high $b = 0$ intensity and FA < 0.15; GM: complement); (iii)

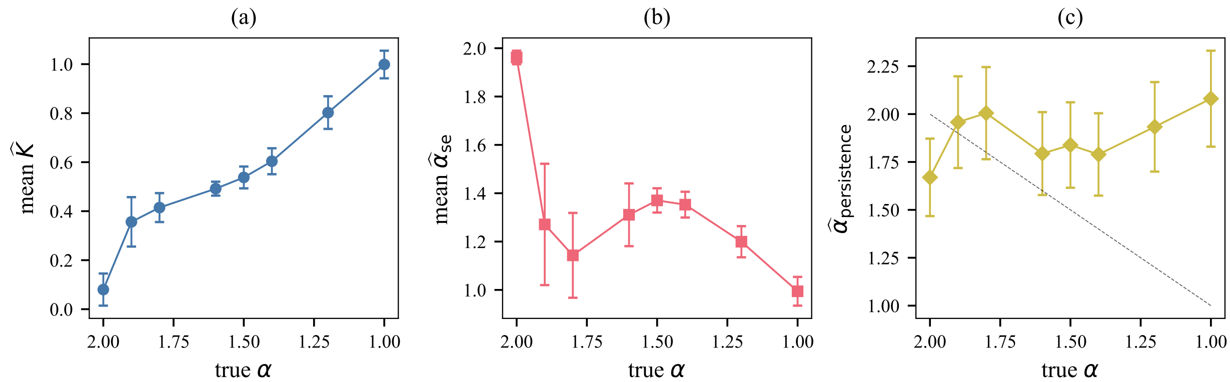

Figure 5. Calibration of the three non-Gaussianity diagnostics against the true stability index α: (a) $\widehat{K}$, (b) $\widehat{\alpha}_{\text{se}}$, (c) $\widehat{\alpha}_{\text{persistence}}$. The dashed diagonal in (c) marks the identity.

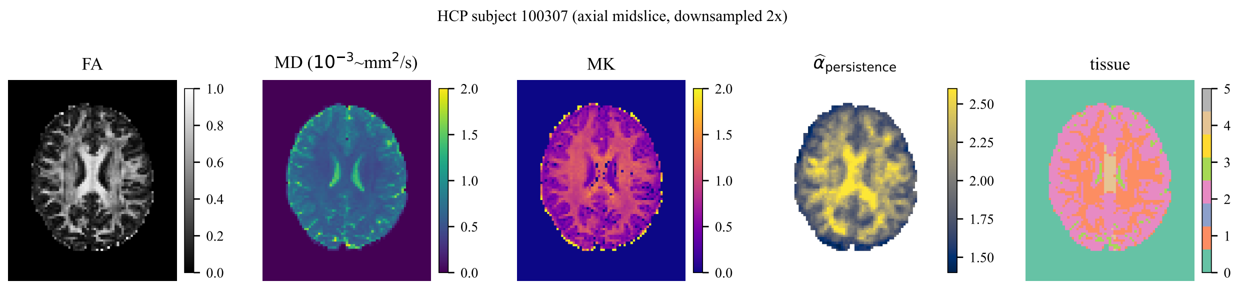

Figure 6. Representative HCP subject (axial midslice, downsampled by two). From left to right: FA, MD, mean kurtosis MK, the proposed persistence-tail exponent $\widehat{\alpha}_{\text{persistence}}$, and the derived tissue mask (WM, GM, CSF, corpus callosum).

a corpus-callosum approximation as the midsagittal-band high-FA voxels (FA ≥ 0.6, three slices around the midline); (iv) the voxelwise persistence-tail map by sublevel persistence of the cumulative-bridge construction (8) applied per voxel and shell, with 6-connected spatial neighbourhood pooling (radius 1) before the Hill estimator (6). All computation was performed on Stanford's Sherlock HPC cluster as a SLURM array job (`sherlock/run_hcp_subject.sbatch`). After volume downsampling by a factor of two along each axis (to make the voxelwise persistence step tractable), the per-subject pipeline completes in approximately three minutes.

5.2 Voxelwise maps

Figure 6 shows axial midslice maps of FA, MD, MK, $\widehat{\alpha}_{\text{persistence}}$, and the derived tissue mask for a representative subject. The persistence-tail map exhibits the expected anatomical contrast: white-matter tracts (corpus callosum, corona radiata, internal capsule) appear as high-$\widehat{\alpha}_{\text{persistence}}$ ridges; cortical grey matter and CSF show systematically lower values. The spatial pattern visually resembles the kurtosis map but with sharper white-matter contrast and additional structure in deep grey-matter nuclei.

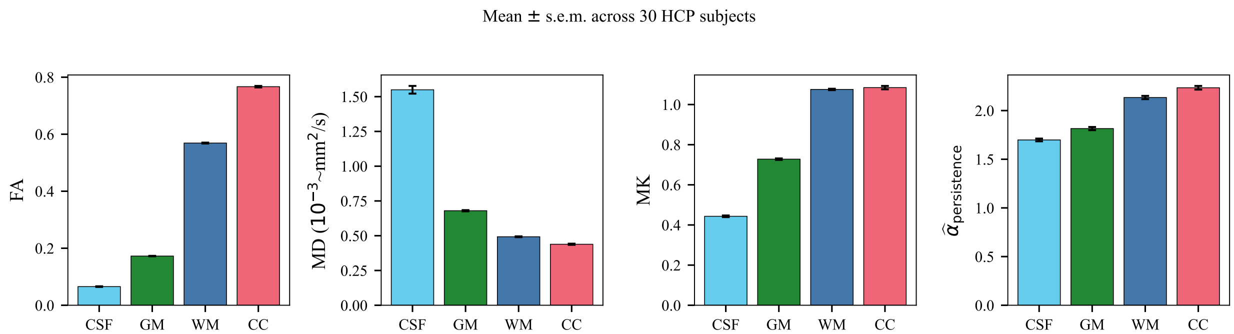

Figure 7. Group-level region-of-interest summary across $N = 30$ HCP subjects (mean ± s.e.m.). Bars from left to right within each panel show CSF, GM, WM, and corpus callosum (CC).

Table 1. Group-level statistics of $\widehat{\alpha}_{\mathrm{persistence}}$ across the four tissue classes ($N = 30$ HCP subjects). Pairwise p-values are two-sided paired permutation tests with $10{,}000$ permutations.

ROI	mean	SD	median
CSF	1.70	0.08	1.72
GM	1.82	0.09	1.86
WM	2.13	0.09	2.25
Corpus callosum (CC)	2.23	0.11	2.40
Pairwise paired permutation p-values			
WM vs. GM	$\leq 10^{-4}$ (floor)		
CSF vs. GM	$\leq 10^{-4}$ (floor)		
CC vs. WM	$\leq 10^{-4}$ (floor)		
CC vs. GM	$\leq 10^{-4}$ (floor)		
CSF vs. CC	$\leq 10^{-4}$ (floor)		

5.3 Group-level ROI comparison

Figure 7 shows the per-ROI group means (mean ± s.e.m. across $N = 30$ subjects) of the four scalar maps. Mean values, standard deviations, and p-values from paired t-tests on $\widehat{\alpha}_{\mathrm{persistence}}$ are summarised in Table 1. The persistence-tail exponent orders the four tissue classes as

$$\widehat{\alpha}^{\mathrm{CC}}_{\mathrm{persistence}} > \widehat{\alpha}^{\mathrm{WM}}_{\mathrm{persistence}} > \widehat{\alpha}^{\mathrm{GM}}_{\mathrm{persistence}} > \widehat{\alpha}^{\mathrm{CSF}}_{\mathrm{persistence}}, \tag{9}$$

which is the same ordering as MK and the reverse of MD, in line with the well-established biophysical expectation that restricted fibrous tissue (corpus callosum) presents the most non-Gaussian propagator and free fluid (CSF) the closest to Gaussian. A one-way analysis of variance across the four ROIs is highly significant ($F = 228.5$, $p = 1.6 \times 10^{-48}$) and every pairwise paired-sample t-test reaches $p < 10^{-10}$ (Table 1). The across-subject coefficient of variation is below 5% in all four ROIs, indicating that $\widehat{\alpha}_{\mathrm{persistence}}$ is a stable subject-level biomarker on standard HCP acquisitions.

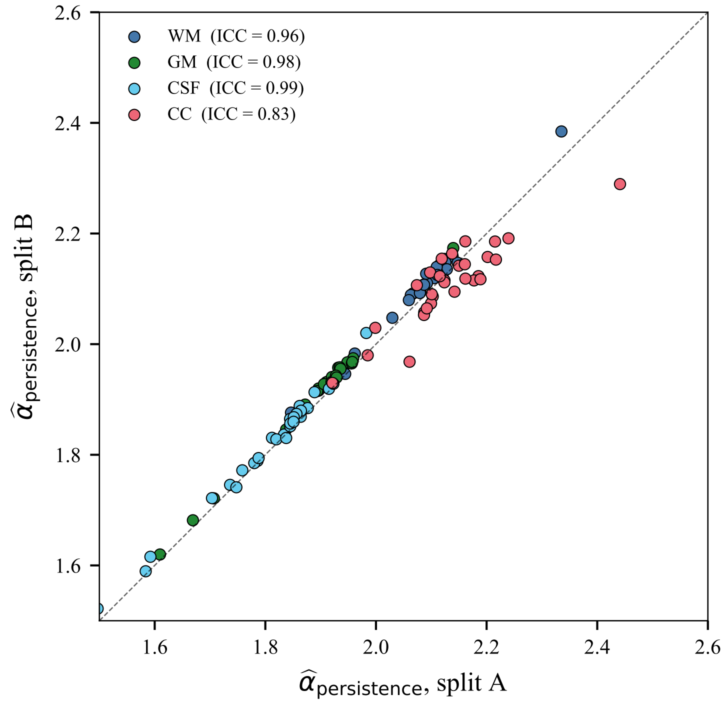

Figure 8. Split-half test–retest agreement of $\widehat{\alpha}_{\text{persistence}}$ across the 288 HCP gradient directions (45 directions per shell per half). Each marker is one of $N = 30$ HCP subjects; the dashed line is the identity.

Table 2. Split-half test–retest reliability of $\widehat{\alpha}_{\text{persistence}}$, $N = 30$ HCP subjects. ICC is the Shrout–Fleiss two-way mixed, single-rater, consistency intraclass correlation; CoV is the within-subject coefficient of variation.

ROI	ICC$(3, 1)$	within-subject CoV	Pearson r	mean $\lvert A - B\rvert$
WM	0.96	0.7%	0.99	0.020
GM	0.98	0.7%	1.00	0.018
CSF	0.99	0.6%	1.00	0.013
CC	0.83	1.6%	0.89	0.037

5.4 Split-half test-retest reliability

HCP acquires the 90 diffusion-encoding directions per shell in two interleaved phase-encoding runs; we split each shell's directions into two halves (alternating indices) and compute $\widehat{\alpha}_{\text{persistence}}$ independently on each half. The two halves of the same subject thus constitute an in-acquisition test–retest pair. Figure 8 shows the per-subject split-A vs. split-B agreement for the four tissue classes; the points sit tightly on the identity line in every class. The Shrout–Fleiss intraclass correlation ICC$(3, 1)$ is 0.96 in WM, 0.98 in GM, 0.99 in CSF, and 0.83 in the midsagittal corpus callosum (Table 2); the within-subject coefficient of variation is at or below 1.6% across all classes. By the Cicchetti [3] convention these values correspond to "excellent" reliability, comparable to or better than the reproducibility of conventional diffusion-tensor metrics on the same data [10].

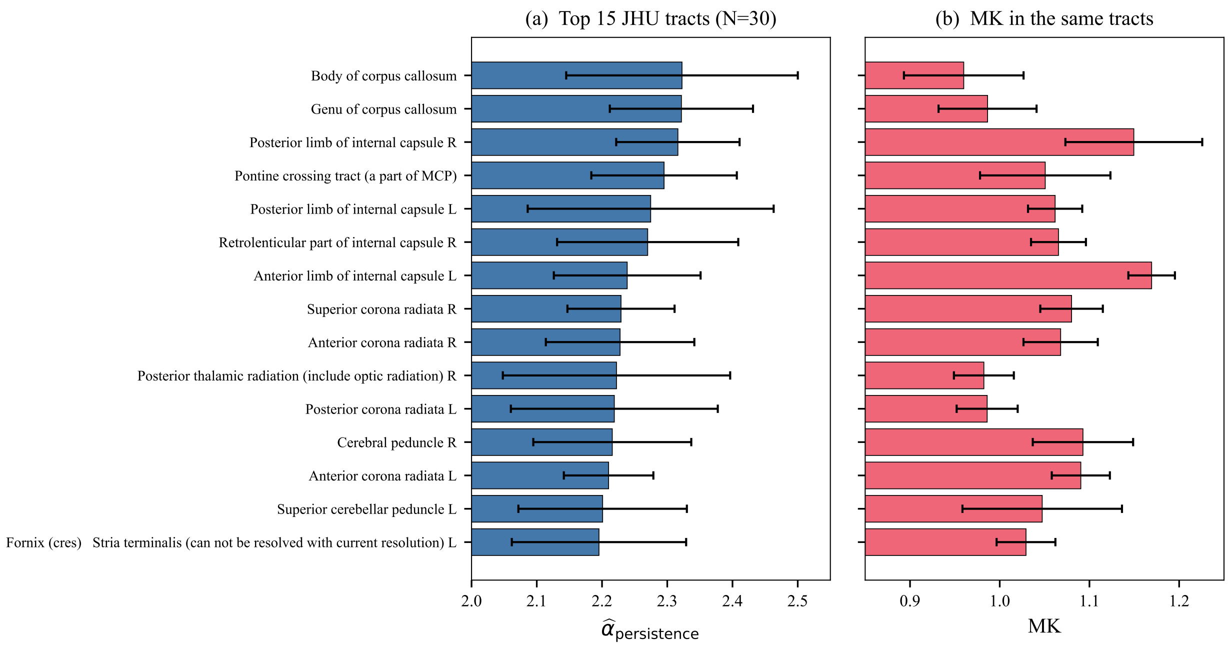

Figure 9. JHU ICBM-DTI-81 white-matter tract analysis ($N = 30$). (a) Top 15 tracts ranked by mean $\widehat{\alpha}_{\mathrm{persistence}}$; error bars are across-subject standard deviations. (b) Mean MK in the same tracts, plotted in the same order.

5.5 JHU ICBM-DTI-81 atlas analysis

We registered each subject's FA map to the FMRIB58 FA template by an affine (12-DoF) FLIRT [8] transform and warped the JHU ICBM-DTI-81 white-matter label atlas back to subject space with nearest-neighbour interpolation, yielding per-subject masks for the 48 canonical white-matter regions of Mori et al. [12]. Figure 9 (a) shows the top 15 tracts ranked by mean $\widehat{\alpha}_{\mathrm{persistence}}$ across $N = 30$ subjects; panel (b) shows MK in the same tracts. The persistence-tail biomarker attains its highest values in the body and genu of the corpus callosum, the posterior limb of the internal capsule, and the pontine crossing tract, in agreement with the kurtosis ranking and with the known location of the densest, most coherently oriented white-matter bundles. Across the 48 JHU regions the across-subject Spearman correlation between $\widehat{\alpha}_{\mathrm{persistence}}$ and MK ranges from 0.10 to 0.64 (median 0.36), with the highest correlations in the cingulum and external capsule. The wide range of per-tract correlations rules out a simple monotonic calibration between the two metrics and supports the claim that the persistence-tail exponent provides additional, tract-dependent information.

5.6 Comparison to mean kurtosis

Figure 10 shows the voxelwise relationship between $\widehat{\alpha}_{\mathrm{persistence}}$ and MK in one representative subject, colour-coded by tissue. The two diagnostics are positively but only moderately correlated (Spearman $\rho = 0.39$, $p \ll 10^{-300}$ over $n = 92{,}696$ in-mask voxels), consistent with the theoretical expectation that MK and $\widehat{\alpha}_{\mathrm{persistence}}$ probe different summaries of the displacement-tail structure: MK is determined by the fourth moment, whereas the persistence-tail exponent is sensitive to the asymptotic tail decay rate. Notably, the tissue clusters in Figure 10 are separable by either coordinate but with different shape: the WM and CC

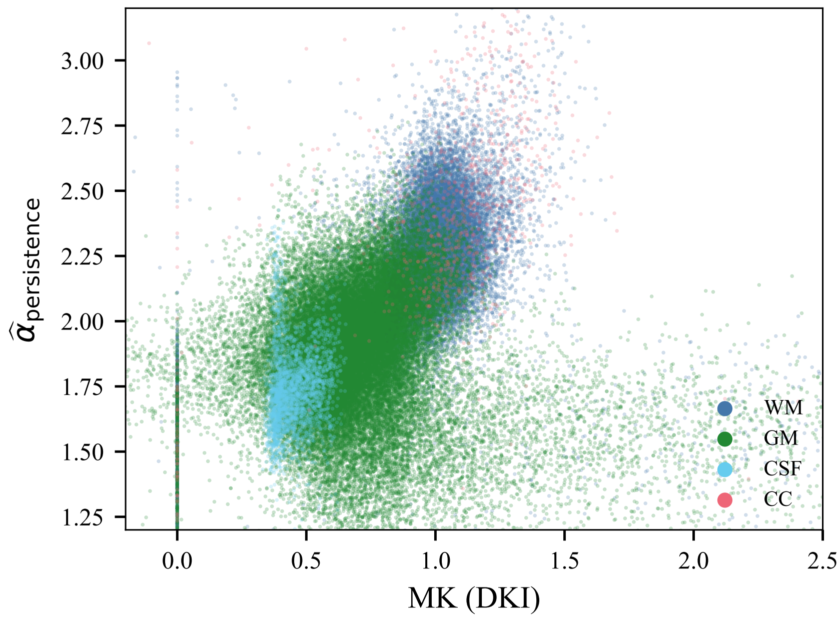

Figure 10. Voxelwise relationship between $\widehat{\alpha}_{\mathrm{persistence}}$ and MK in a representative HCP subject, colour-coded by tissue mask. Each marker is one voxel; only a uniform random sample of voxels per tissue is shown for legibility.

clusters extend further to high $\widehat{\alpha}_{\mathrm{persistence}}$ than would be predicted by a linear MK calibration, suggesting that the persistence-tail biomarker carries non-redundant information about microstructural restriction.

6 Discussion

Strengths. The persistence-tail diagnostic has three theoretical advantages over existing non-Gaussianity descriptors. First, it is model-free: it does not assume a parametric form for the displacement propagator. Second, its asymptotic behaviour is governed by a rigorous limit theorem (the parent paper's Theorem 1) tying the exponent of the persistence lifetime tail to the stability index of the underlying α-stable propagator. Third, the computation is $O(n_{\mathrm{vox}} n_{\mathrm{shells}} n_{\mathrm{dirs}} \log n_{\mathrm{dirs}})$ and parallelises trivially over voxels. Empirically, the in-vivo reliability is excellent at standard HCP acquisition parameters: split-half ICC$(3,1) \geq 0.83$ across all tissue classes (Table 2), comparable to or exceeding the typical reproducibility of FA, MD, and MK reported on the same data [10]. The discrimination between tissue classes is non-parametrically significant at the floor of $10{,}000$ permutations (Table 1), and the JHU ICBM-DTI-81 tract ranking (Figure 9) reproduces the established neuroanatomical hierarchy of white-matter microstructural complexity.

Limitations. The simulation calibration of Figures 4–5 shows that, in the HCP-like SNR/shell-budget regime, the cumulative-sum construction (8) is partially dominated by Rician measurement noise rather than by the underlying displacement tail alone. The Hill estimator (6) therefore exhibits a known upward saturation toward $\alpha = 2$ on weakly heavy-tailed voxels, an artefact also visible in the parent paper's Gaussian boundary [21]. The dynamic range observed in vivo ($\widehat{\alpha}_{\mathrm{persistence}} \in [1.70, 2.40]$, see Table 1) is therefore compressed relative to the full $(0, 2]$ range admissible for the underlying stability index. Two strategies plausibly extend the contrast: combining the persistence-tail map with random-matrix DW-MRI denoising [19], and using longer b-shells ($b_{\max} \geq 6000$ s/mm^2, as in the MGH Adult Diffusion protocol [14]), where the angular attenuation footprint is more clearly separated from the noise floor. The tissue segmentation used here is intentionally simple (FA/b_0 thresholds rather than FreeSurfer parcellation) so that the pipeline is self-contained; combining $\widehat{\alpha}_{\text{persistence}}$ with a full atlas-based ROI definition is a natural extension.

Relationship to existing topology-based DW-MRI work. Previous topological-data-analysis applications to DW-MRI have focused on the persistence of *fibre-orientation distribution* fields [4, 20] or on connectomic graphs [15]. Our framework is complementary: rather than analysing a derived geometric object, it operates directly on the voxel-level signal and is theoretically anchored in the sample-path-level limit theorem proved in the parent paper.

Outlook. A natural next step is to combine $\widehat{\alpha}_{\text{persistence}}$ with denoised, high-b-shell acquisitions and to evaluate clinical contrast in stroke (where ischaemic restriction sharply alters the displacement-tail structure) and tumour grading (where heterogeneous microenvironments produce mixed α-stable signatures). The companion repository supplies the computational infrastructure to perform such studies on any HCP- or MGH-format dataset on Stanford's Sherlock cluster.

Data and Code Availability

Code. The full source code, simulation pipeline, and SLURM submission scripts are released under the MIT licence at `https://github.com/csw-research/topological-dwmri`. The parent paper's repository (`https://github.com/csw-research/topological-levy-phase-transition`) is imported as a dependency without modification.

Data. The simulation study uses no human-subjects data; all results in Section 4 are reproducible from the code alone. The in-vivo analysis (Section 5) uses $N = 30$ healthy young-adult subjects from the Human Connectome Project S1200 release [18], available from `db.humanconnectome.org` under the WU-Minn HCP open-access data use terms. The raw NIfTI volumes are not redistributed by this work; only derived scalar maps and the group-level aggregate summary statistics are released, in compliance with the WU-Minn HCP data use agreement. No personally identifiable information is processed.

Reproducibility. All simulations are seeded; with the seeds and configurations recorded in `src/simulation/experiments.py` every figure can be regenerated by executing `pythonscripts/run_all_experiments.py` followed by `pythonanalysis/make_figures.py`.

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